Neural Networks: Introduction

Video Lesson

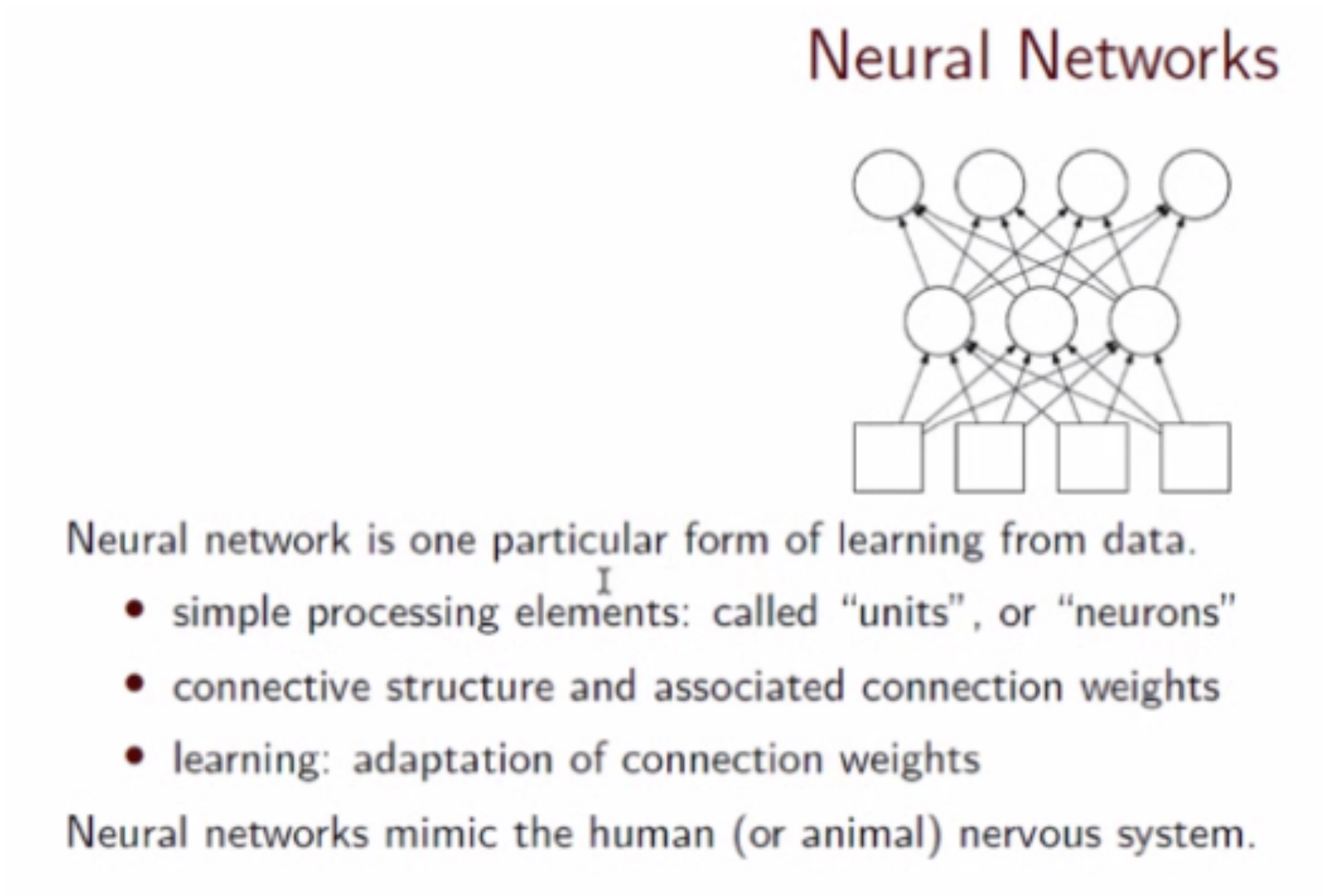
Duration: 00:10:50

Slide Deck

M12S03 (https://canvas.tamu.edu/courses/430127/files/92570696?wrap=1)

Lesson Transcript

In this and several following lectures, we will look into neural networks.



The neural network is one particular form of learning from data. It consists of simple processing elements. These are called units or neurons, and they have a connective structure like these, and each of these connections is associated with a connection weight. Learning is based on the adaptation of these connection rate values.

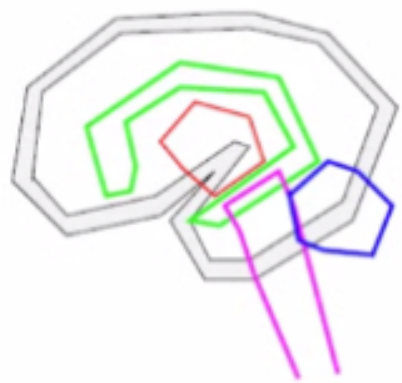
At an abstract level, neural networks mimic the human nervous system.

Many Faces of Neural Networks

- Abstract mathematical/statistical model
- Optimization algorithm
- Pattern recognition algorithm
- Tools for understanding the function of the brain
- Robust engineering application

There are many faces of neural networks. We can consider them as abstract mathematical and statistical models. Alternatively, we can treat them as some form of optimization algorithm or a pattern recognition algorithm. We can also consider them as tools for understanding the function of the brain or as a robust engineering application.

The Central Nervous System



- Cortex: thin outer sheet where most of the neurons are.
- Sub-cortical nuclei: thalamus, hippocampus, basal ganglia, etc.
- Midbrain, pons, and medulla, connects to the spinal cord.
- Cerebellum (hind brain, or small brain)

Here is the central nervous system, which is the main inspiration for neural networks. There is a cortex, a thin outer sheet where most of the neurons are. Sub-cortical nuclei, such as the thalamus, the hippocampus, and basal ganglia. There are also the midbrain, pons, and medulla. These connect through the spinal cord and the cerebellum, the hindbrain.

Function of the Nervous System

Function of the nervous system:

- Perception
- Cognition
- Motor control
- Regulation of essential bodily functions

There are various functions of the nervous system, perception, cognition, motor control, regulation of essential bodily functions, and so on, and we will, later on, see how these neural networks or deep learning algorithms derived from neural networks can be used to perform these different tasks.

The Central Nervous System: Facts

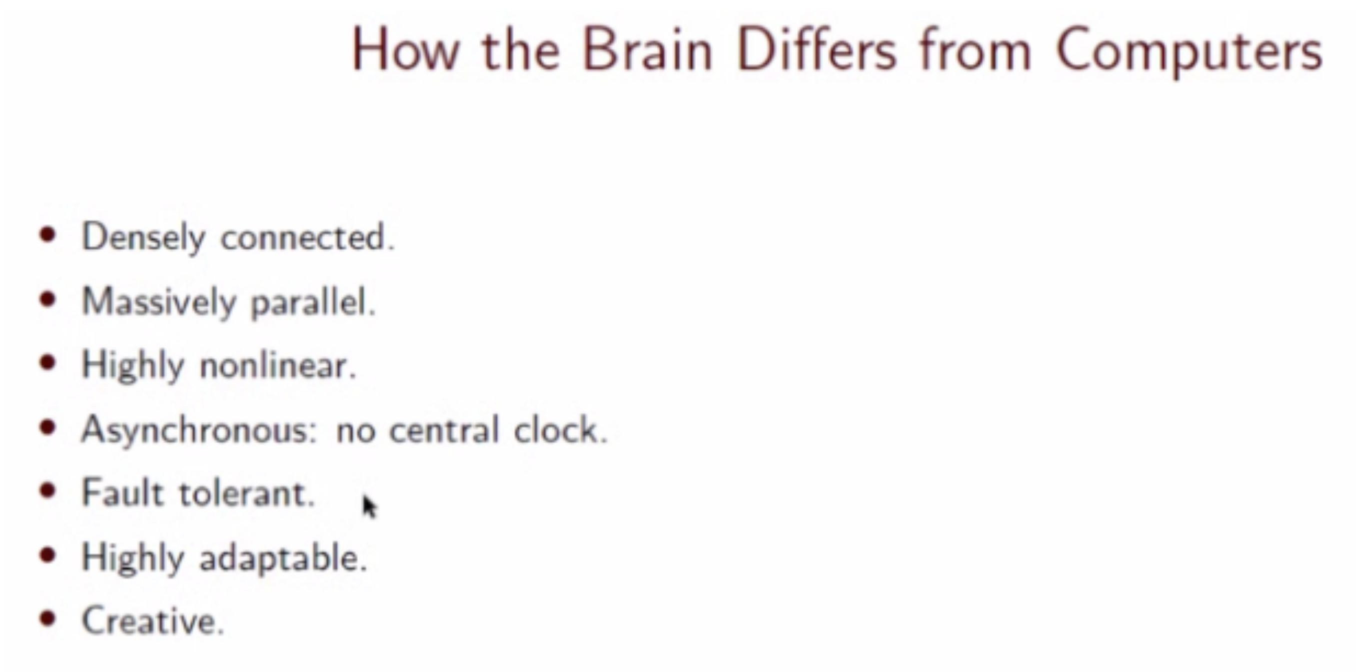
Facts: human brain

- Neocortex Thickness: 1.6mm
- Neocortex Area: 2,500 cm^2 (about 2.5 ft^2: Peters and Jones, 1984)
- Neurons: 86 billion (86×10^9: Frederico Azevedo el al. 2009), 10 billion in the neocortex (Shephered, 1998).
- Connections: 150 trillion (150^{12}: Pakkenberg et al. 1997, 2003). 60 trillion in the neocortex (Shephered, 1998).
- Connections per neuron: $\sim 1,800$ (from above)
- Energy usage per operation: 10^{-16}J (compare to 10^{-6}J in modern computers)

Unless specified, the numbers are from *Neural networks: a comprehensive foundation* by Simon Haykin (1994), and *Foundations of Vision* by Brian Wandell (1995).

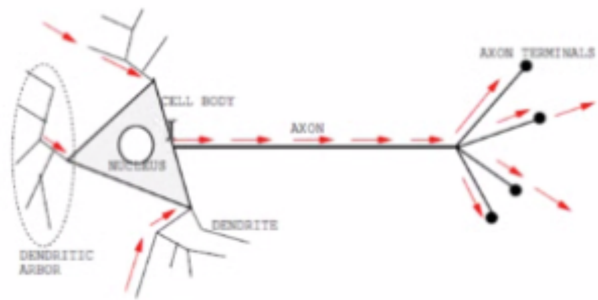
Here are some facts about the human brain. The neocortex, the surface of the brain that has all of these neurons packed in there, the thickness is about 1.6 millimeters. The total area, if unfolded or flattened the surface area is 2,500 square centimeters, or that is about 2.5 square feet, and there are about 86 billion neurons in it and about 150 trillion connections. That is a huge number of neurons and connections.

The connection per neuron is about 1,800. Thus, each neuron connects to about 1,800 other neurons. One of the striking facts about the brain is that it is very energy efficient. It is over 10 orders of magnitude more efficient than modern computers.



The brain differs from conventional computers. It is densely connected. It is massively parallel. It is highly nonlinear. And it is asynchronous. That is, there is no central clock. It is fault-tolerant. If one is damaged to part of one's brain, one can still survive. And it is highly adaptable and creative.

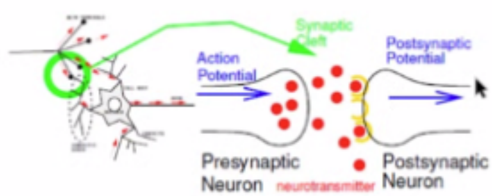
Neurons: Basic Functional Unit of the Brain



- Dendrites receive input from upstream neurons.
- Ions flow in to make the cell positively charged.
- Once a firing threshold is reached, a spike is generated and transmitted along the axon.
- Axon terminals release neurotransmitters to relay the signal to the downstream neurons.

Neurons are the basic functional units of the brain. It has the cell body and dendrites where it receives the input, and it has the axon where the spikes, the neuron's output, get transmitted to other neurons.

Propagation of Activation Across the Synapse

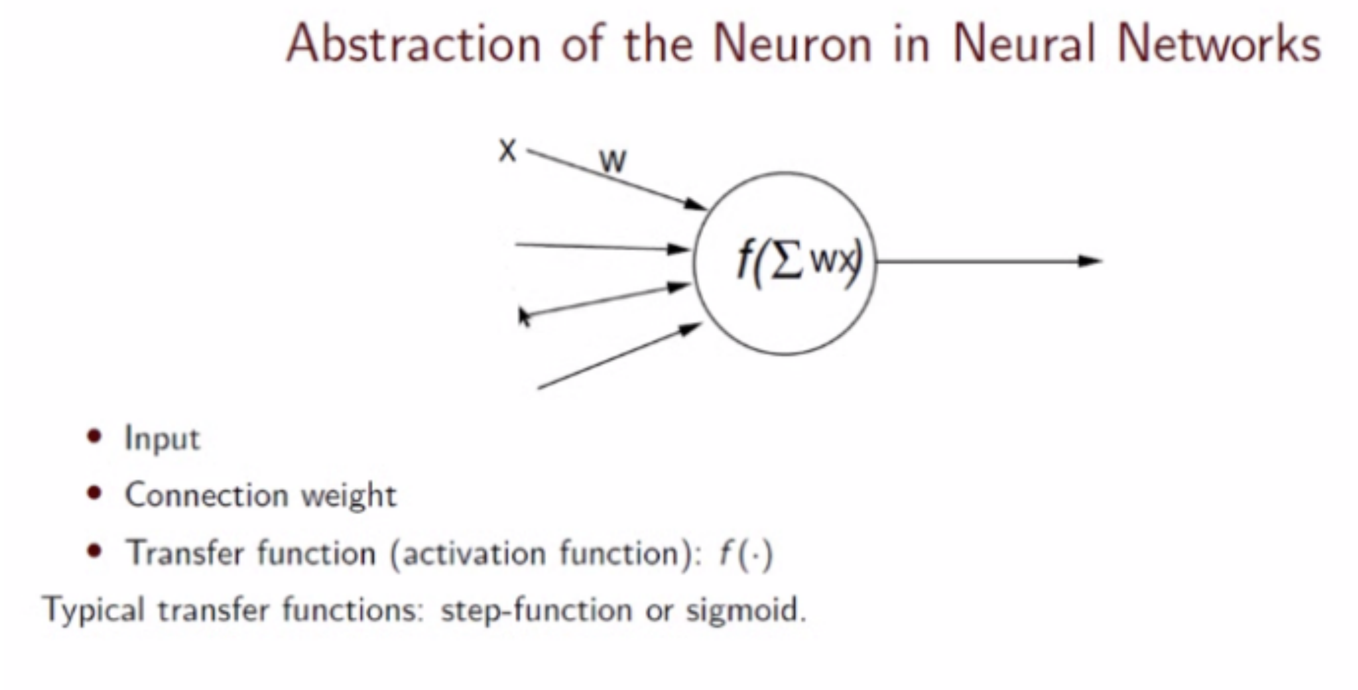


1. Action potential reaches axon terminal.
2. Neurotransmitters are released into the synaptic cleft and bind to postsynaptic cell's receptors.
3. Binding allows ion channels to open (Na^+), and Na^+ ions flows in and makes the postsynaptic cell depolarize.
4. Once the membrane voltage reaches the threshold, an action potential is generated.

Lesson: neural activity propagation has a very complex cellular/molecular mechanism.

The interface between the output side of the previous neuron and the input side of the next neuron is called the synapse, and there is a gap. On the left-hand side is the pre-synaptic neuron, the input side. On the right-hand side is the cell membrane of the postsynaptic neuron, the output side. These electrical signals get transmitted through the axon, and when they are received at the end of this presynaptic neuron, then neurotransmitters are released into this gap, and they bind with the receptors on the output side or the receiving end, and these trigger certain channels to open, and ions to flow to generate more current.

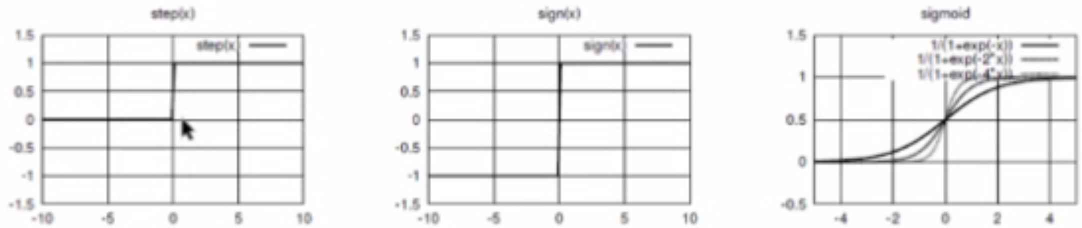
Of course, you do not have to memorize any of these for the exam, but one thing you want to realize is that what our brain does is a very complex electrochemical process.



All of that complexity is abstracted to a single value or a single real numbered value in an artificial neuron network.

Here is an artificial neuron or a unit that takes in several input values, and it has a connection rate on each incoming connection where the product of the input value and the weight value is added, and that is finally passed through something called the transfer function or the activation function, and that would be the final output value of this neuron or the unit.

Typical Activation Functions



- $Step_t(x) = 1$ if $x \geq t, 0$ if $x < t$
- $Sign(x) = +1$ if $x \geq 0, -1$ if $x < 0$
- $Sigmoid(x) = \frac{1}{1+e^{-x}}$

Note that $Step_t(x) = Step_0(x - t)$, which we will simply call $Step(x - t)$.

$$Step_0(x-t) = 1 \text{ if } \underbrace{x - t \geq 0}_{\text{same as } x \geq t}, 0 \text{ if } \underbrace{x - t < 0}_{\text{same as } x < t}$$

Here are some typical activation functions. The first is the step function, the next is the sign function, and the last one is the sigmoid function.

The step function is 1 if the input value is greater than or equal to the threshold, *t*. The plot here shows when that threshold *t* is 0.

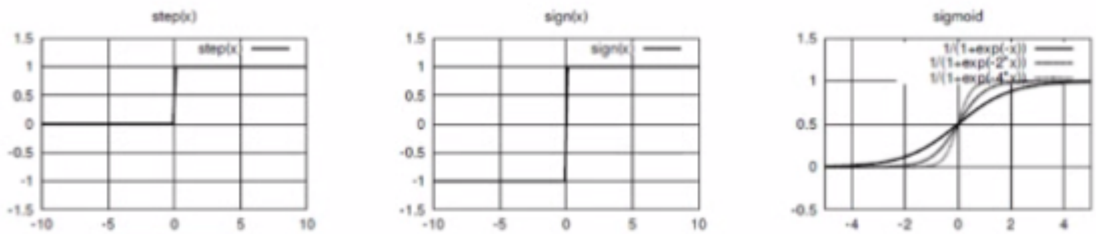
If that input value *x* is less than the threshold *t*, then the output is 0.

The sign function is very similar to the step function, whereas here, the output is -1 or +1.

Finally, we have the sigmoid function, which generates an output between 0 and 1 and a graded value in between. Sigmoid *x* is 1 over 1 plus the exponential of -*x*.

One interesting thing about this step function is that we can redefine step *t* of *x*, where the threshold is *t* into step function with 0 as the threshold if we incorporate this threshold *t* into the input itself as *x* - *t*.

Typical Activation Functions



- $Step_t(x) = 1$ if $x \geq t$, 0 if $x < t$
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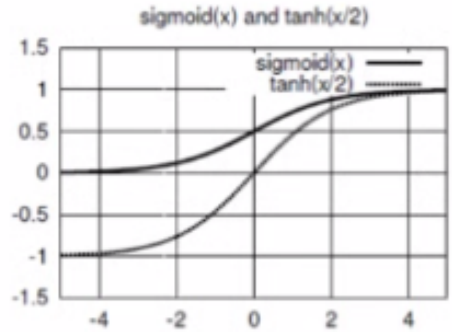
$$Step_0(x - t) = 1 \text{ if } \underbrace{x - t \geq 0}_{\text{same as } x \geq t}, 0 \text{ if } \underbrace{x - t < 0}_{\text{same as } x<t} = Step_t(x)$$

What we can do is plug in 0 to the threshold *t* and the input *x* - *t* into here. Then this becomes that. This output is 1 if *x* - *t* is greater than or equal to 0 because this threshold set here is to be 0. And 0 if *x* - *t*, this input, is less than 0. But just by adding both sides with *t*, we get *x* greater than equal to *t*. Adding both sides of this inequality with *t* will get us *x* less than *t*.

Because these are the same, this basically defines step *t* of *x*. Hence, step *t* of *x* equals step 0 of *x* - *t*.

More Activation Functions: $tanh(\frac{x}{2})$



- $Sigmoid(x) = \frac{1}{1+e^{-x}}$
- $tanh(\frac{x}{2}) = \frac{1-e^{-x}}{1+e^{-x}}$

There are also variations of the sigmoid function, which ranges between 0 and 1. We can change this to the hyperbolic tangent function, or *tanh*, and we get this formula. The resulting function looks like this, which is basically a sigmoid that ranges between -1 and 1.

Classification of Neural Networks

Teacher exists?

- Supervised (with teacher): perceptrons, backpropagation network, etc.
- Unsupervised (no teacher): self-organizing maps, PCA, k-means, etc.

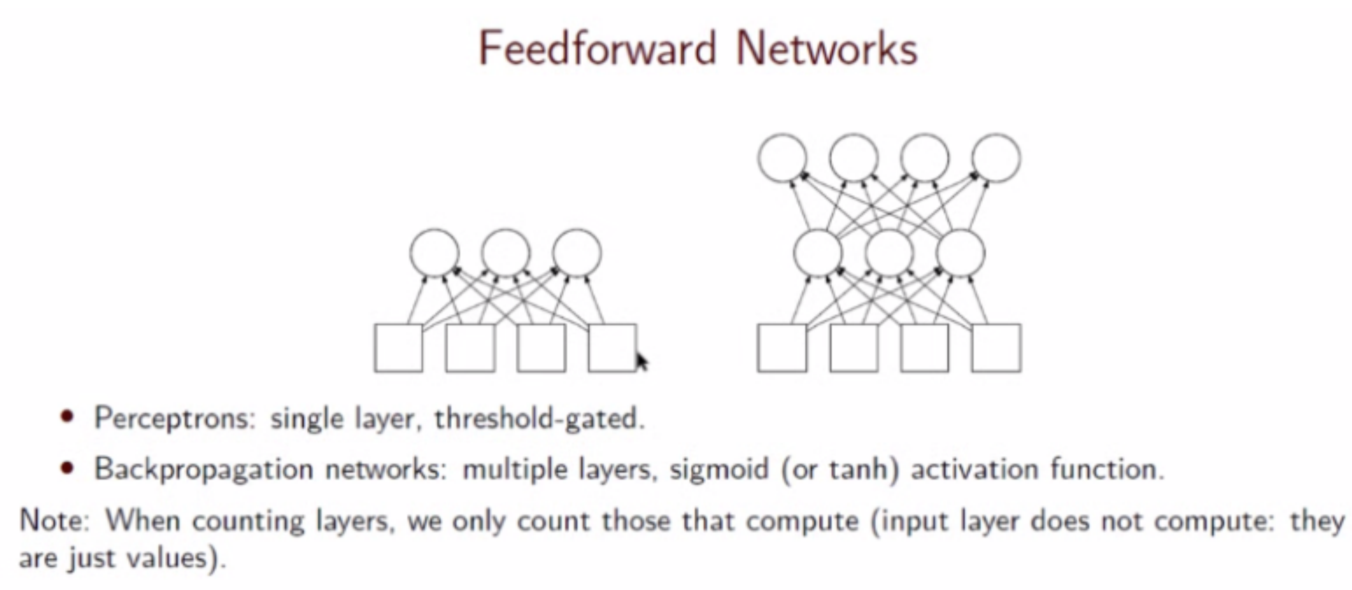
Recurrent connections?

- Feed-forward: perceptrons, backpropagation network, etc.
- Recurrent: Hopfield network, Boltzmann machines, SRN (simple recurrent network), LSTM, GRU, etc.

There are several ways to classify neural networks into different types. We can categorize them based on two criteria. Is there a teacher signal or not? Are there recurrent connections, connections that form a loop or not? For this condition, is the teacher there to provide us with feedback?

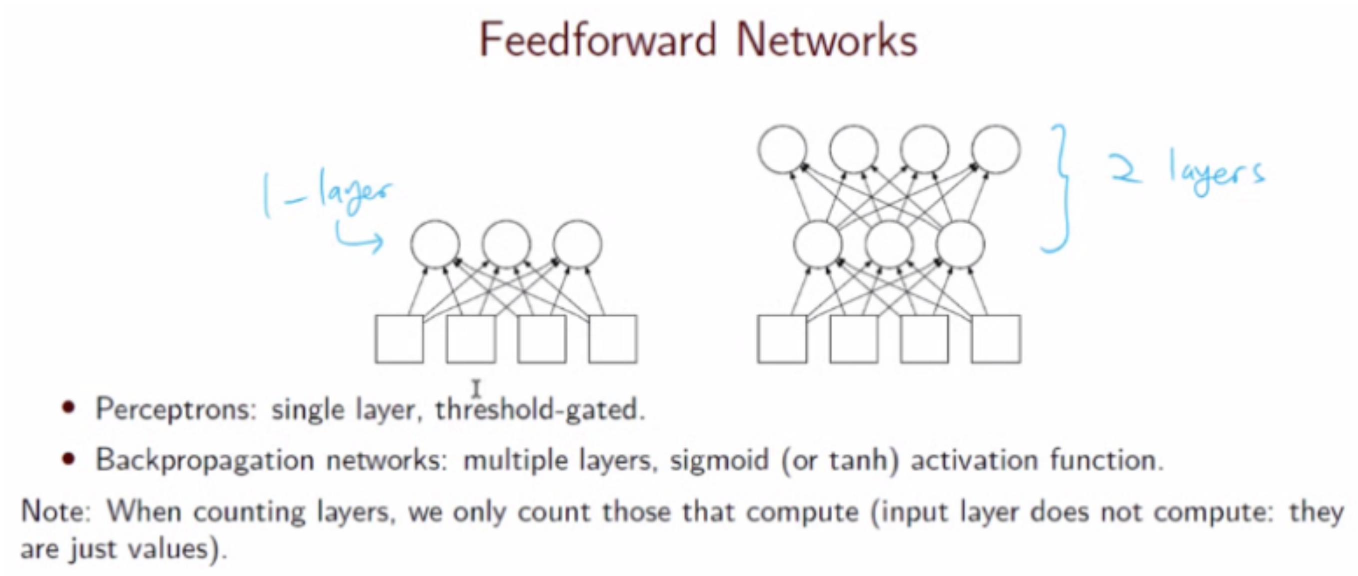
If we have this teacher signal, then it is a supervised neural network, and the examples include perceptrons, backpropagation networks, multilayer perceptrons, and so on. Unsupervised neural networks include self-organizing maps, etc. Here again, it is basically just a collection of inputs or input points that come in as the input in the dataset.

If we do not have recurrent connections, it is called a Feed-forward Neural Network. We will mostly focus on these in this particular lecture and the several following ones. If we have recurrent connections, the examples include the Hopfield network, Boltzmann machines, simply recurrent networks, and networks that include something called LSTM, long short-term memory, GRU, and so on.



Again, in the next few lectures, we will focus on Feed-forward Neural Networks. It could be a single layer like this. We have a single layer of computing units. These are input units, and they do not perform any computation. They are just real-number values. A network that looks like this is called a single-layer neural network.

When we have a threshold-gated output, meaning that this is basically a step function, then these are called perceptrons, and we have backpropagation networks. These backpropagation networks can have multiple layers.



As we can see, using the same reasoning, this network has two computing layers. Hence it is called a two-layer backpropagation network or two-layer multi-layer perceptron.

In the following lecture, we will look at this perceptron in more detail.